Practice Problems

A physics student wishes to repeat Fizeau's experiment for measuring the speed of light. If he uses a toothed wheel containing 1440 teeth and his distant mirror is located in a laboratory window across the college campus 412.60 m away, how fast must his wheel be rotated if the returning light pulses show the first maximum intensity?

A beam of light passes through a block of glass 10.0 cm thick, then through water for a distance of 30.5 cm, and finally through another block of glass 5.0 cm thick. If the refractive index of both pieces of glass is 1.5250 and of water is 1.3330, find the total optical path.

In studying the refraction of light Kepler arrived at a refraction formula

$$\phi = \frac{\phi'}{1 - k \sec \phi'} \qquad \text{where} \qquad k = \frac{n' - 1}{n'}$$

n' being the relative index of refraction. Calculate the angle of incidence ϕ for a piece of glass for which $n' = 1.7320$ and the angle of refraction $\phi' = 32.0°$ according to (*a*) Kepler's formula and (*b*) Snell's law. Note that $\sec \phi' = 1/(\cos \phi')$.

A ray of light in air enters the center of one face of a prism at an angle making 55.0° with the normal. Traveling through the glass, the ray is again refracted into the air beyond. Assume the angle between the two prism faces to be 60.0° and the glass to have a refractive index of 1.650. Find the deviation of the ray (*a*) at the first surface and (*b*) the second surface. Find the total deviation (*c*) by calculation and (*d*) graphically.